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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 31st August Morning Shift

JEE Main Question Paper with Answer Key

31 Aug 2021	90	18	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	21 questions
Mathematics	24 questions
Physics	27 questions
Out of syllabus	18 questions
Answer key	at the end

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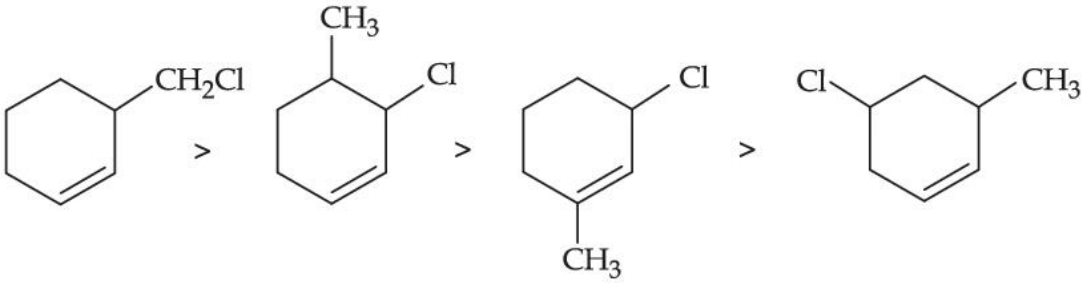
Chemistry 21 questions

Q1. MCQ (Single Correct Answer)

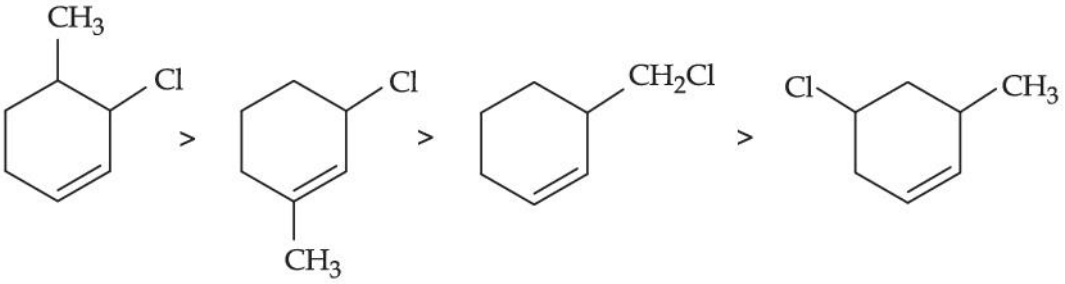
Open solution

The correct order of reactivity of the given chlorides with acetate in acetic acid is :

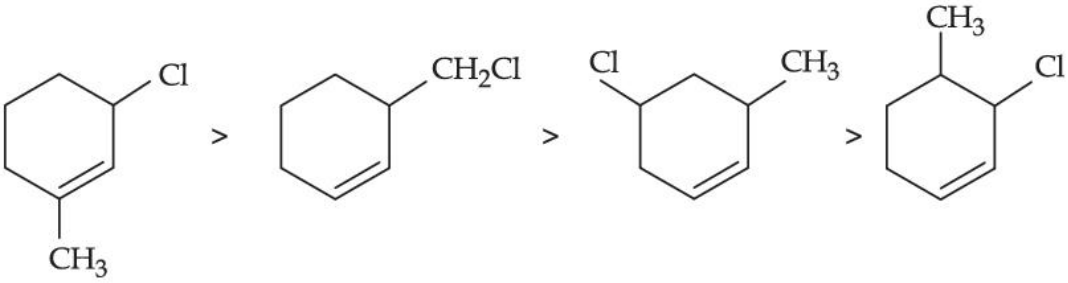
(A)



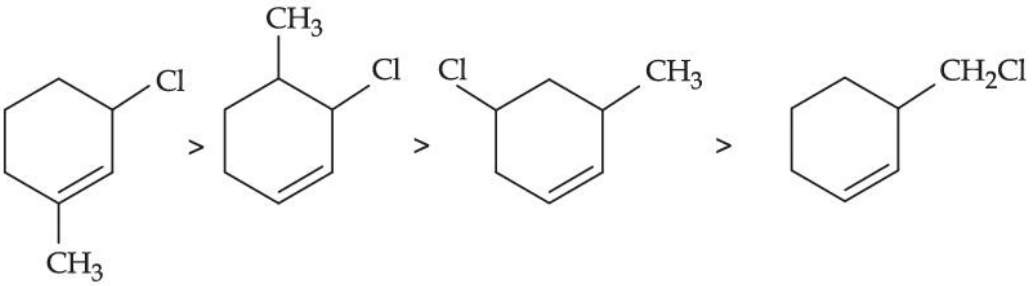
(B)



(C)



(D)



Q2. MCQ (Single Correct Answer)

Open solution

In the structure of the dichromate ion, there is a :

(A) linear symmetrical Cr-O-Cr bond.

(B) non-linear symmetrical Cr-O-Cr bond.

(C) linear unsymmetrical Cr-O-Cr bond.

(D) non-linear unsymmetrical Cr-O-Cr bond.

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Q3. MCQ (Single Correct Answer)

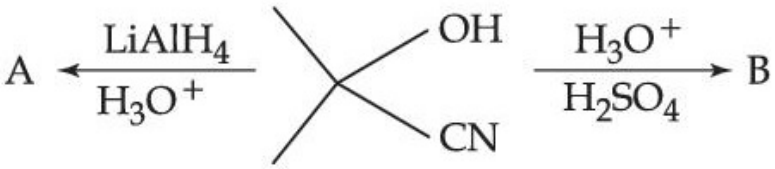
Which one of the following compounds contains β-C_1-C_4 glycosidic linkage?

(A) Lactose (B) Sucrose (C) Maltose (D) Amylose

Open solution

Q4. MCQ (Single Correct Answer)

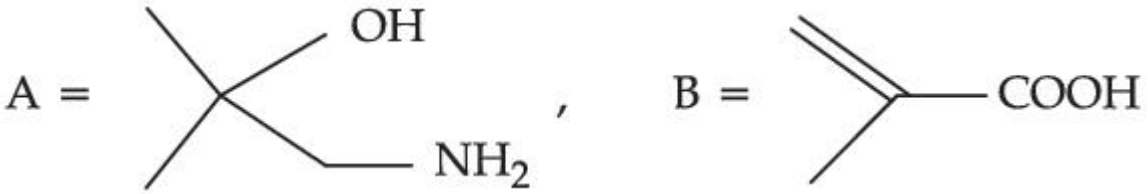
The major products A and B in the following set of reactions are :



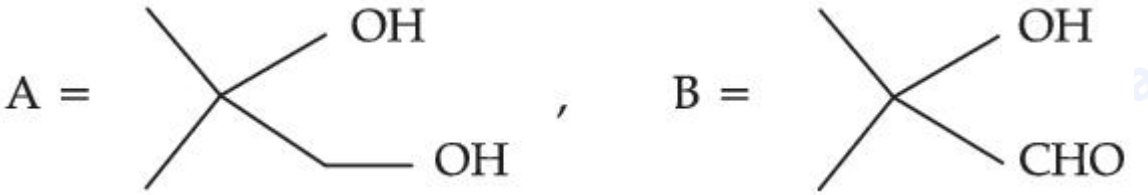
(A)



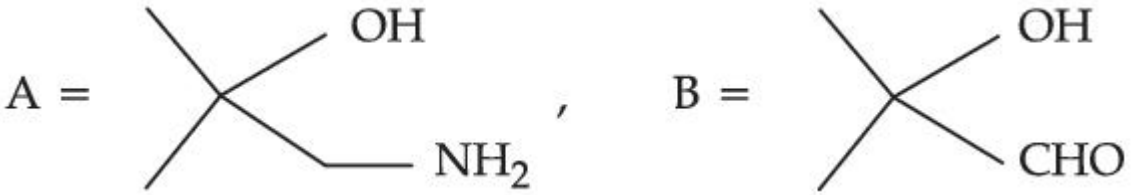
(B)



(C)



(D)



Q5. MCQ (Single Correct Answer)

Which one of the following lanthanides exhibits +2 oxidation state with diamagnetic nature? (Given Z for Nd = 60, Yb = 70, La = 57, Ce = 58)

(A) Nd (B) Yb (C) La (D) Ce

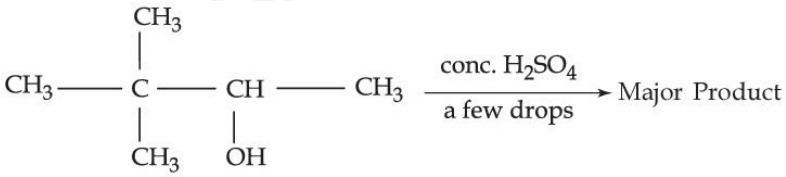
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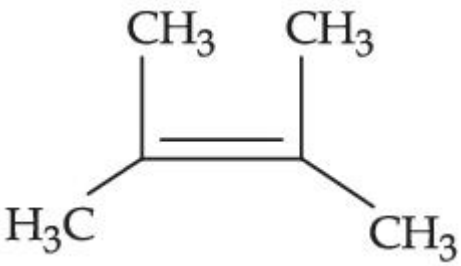
Q6. MCQ (Single Correct Answer)

Open solution

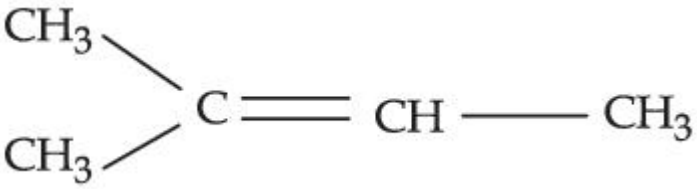
The major product formed in the following reaction is :



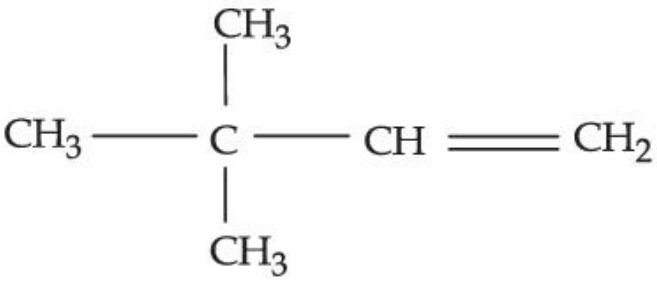
(A)



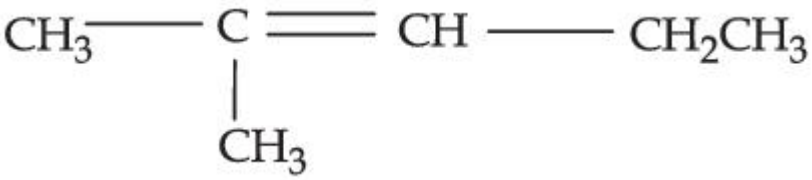
(B)



(C)



(D)



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Q7. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Treatment of bromine water with propene yields 1-bromopropan-2-ol.

Reason (R) : Attack of water on bromonium ion follows Markovnikov rule and results in 1-bromopropan-2-ol.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(B) (A) is false but (R) is true

(C) Both (A) and (R) are true and (R) is the correct explanation of (A)

(D) (A) is true but (R) is false

Q8. MCQ (Single Correct Answer)

Open solution

The denticity of an organic ligand, biuret is :

(A) 2 **(B)** 4 **(C)** 3 **(D)** 6

Q9. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Metallic character decreases and non-metallic character increases on moving from left to right in a period.

Reason (R) : It is due to increase in ionisation enthalpy and decrease in electron gain enthalpy, when one moves from left to right in a period.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) (A) is false but (R) is true.

(B) (A) is true but (R) is false

(C) Both (A) and (R) are correct and (R) is the correct explanation of (A)

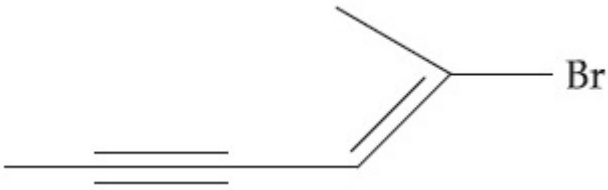
(D) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

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Q10. MCQ (Single Correct Answer)

Open solution

Choose the correct name for compound given below :



(A) (4E)-5-Bromo-hex-4-en-2-yne

(B) (2E)-2-Bromo-hex-4-yn-2-ene

(C) (2E)-2-Bromo-hex-2-en-4-yne

(D) (4E)-5-Bromo-hex-2-en-4-yne

Q11. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R) :

Assertion (A) : A simple distillation can be used to separate a mixture of propanol and propanone.

Reason (R) : Two liquids with a difference of more than 20° in their boiling points can be separated by simple distillations.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) (A) is false but (R) is true.

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) Both (A) and (R) are correct and (R) is the correct explanation of (A).

Q12. MCQ (Single Correct Answer)

Open solution

Which one of the following 0.10 M aqueous solutions will exhibit the largest freezing point depression?

(A) hydrazine **(B)** glucose **(C)** glycine **(D)** $KHSO_4$

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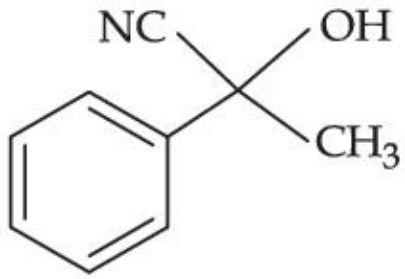
Q13. MCQ (Single Correct Answer)

Open solution

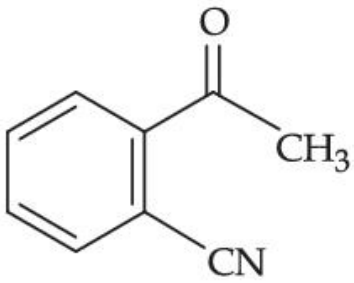
The structure of product C, formed by the following sequence of reactions is :

$$CH_3COOH + SOCl_2 \longrightarrow A \xrightarrow[AlCl_3]{\text{Benzene}} B \xrightarrow[^{-}OH]{KCN} C$$

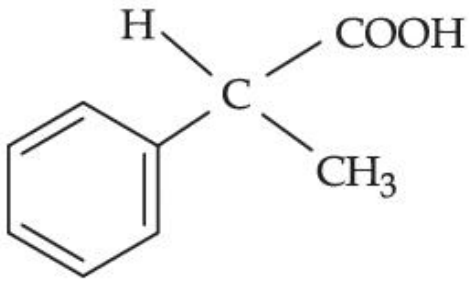
(A)



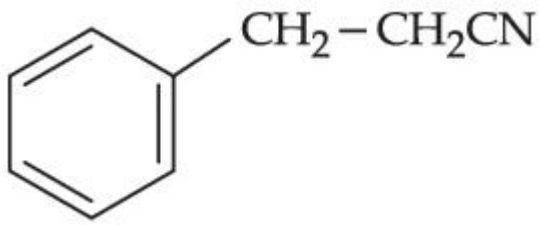
(B)



(C)



(D)



Q14. Numerical

Open solution

Consider the following cell reaction :

$$Cd_{(s)} + Hg_2SO_{4(s)} + \frac{9}{5}H_2O_{(l)} \rightleftharpoons CdSO_4.\frac{9}{5}H_2O_{(s)} + 2Hg_{(l)}$$

The value of E^0_{cell} is 4.315 V at 25°C. If $\Delta H^\circ = -825.2$ kJ mol^{-1}, the standard entropy change ΔS° in J K^{-1} is _____. (Nearest integer) [Given : Faraday constant = 96487 C mol^{-1}]

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Q15. Numerical

The molarity of the solution prepared by dissolving 6.3 g of oxalic acid ($H_2C_2O_4.2H_2O$) in 250 mL of water in mol L^{-1} is $x \times 10^{-2}$. The value of x is _____. (Nearest integer) [Atomic mass : H : 1.0, C : 12.0, O : 16.0]

Open solution

Q16. Numerical

The total number of reagents from those given below, that can convert nitrobenzene into aniline is _____. (Integer answer)

Open solution

I. Sn – HCl

II. Sn – NH_4OH

III. Fe – HCl

IV. Zn – HCl

V. H_2 – Pd

VI. H_2 – Raney Nickel

Q17. Numerical

For a first order reaction, the ratio of the time for 75% completion of a reaction to the time for 50% completion is _____. (Integer answer)

Open solution

Q18. Numerical

The number of hydrogen bonded water molecule(s) associated with stoichiometry $CuSO_4.5H_2O$ is _____.

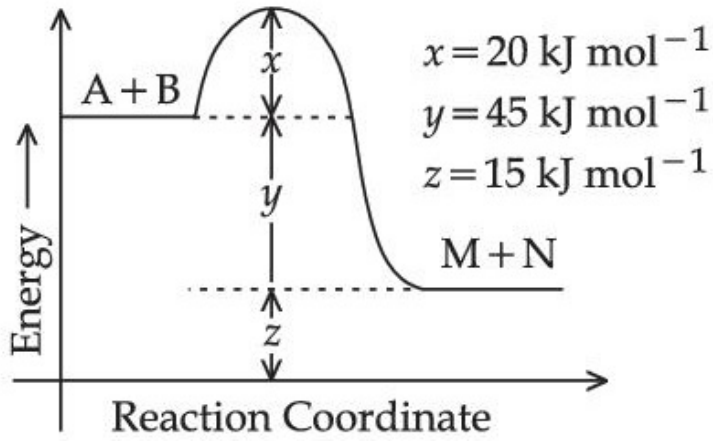
Open solution

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Q19. Numerical

According to the following figure, the magnitude of the enthalpy change of the reaction

A + B $\rightarrow$ M + N in kJ mol^{-1} is equal to _____. (Integer answer)



Open solution

Q20. Numerical

Ge(Z = 32) in its ground state electronic configuration has x completely filled orbitals with $m_l = 0$. The value of x is _____.

Open solution

Q21. Numerical

A_3B_2 is a sparingly soluble salt of molar mass M (g mol^{-1}) and solubility x g L^{-1}. The solubility product satisfies $K_{sp} = a\left(\frac{x}{M}\right)^5$. The value of a is _____. (Integer answer)

Open solution

Mathematics 24 questions

Q22. MCQ (Single Correct Answer)

The number of real roots of the equation

$e^{4x} + 2e^{3x} - e^x - 6 = 0$ is :

(A) 2 (B) 4 (C) 1 (D) 0

Open solution

Q23. MCQ (Single Correct Answer)

Let f be a non-negative function in [0, 1] and twice differentiable in (0, 1). If $\int_0^x \sqrt{1-(f'(t))^2}dt = \int_0^x f(t)dt, 0 \le x \le 1$ and f(0) = 0, then $\lim_{x\to 0} \frac{1}{x^2}\int_0^x f(t)dt$:

(A) equals 0 (B) equals 1 (C) does not exist (D) equals $\frac{1}{2}$

Open solution

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Q24. MCQ (Single Correct Answer)

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|2\vec{a}+3\vec{b}\right| = \left|3\vec{a}+\vec{b}\right|$ and the angle between $\vec{a}$ and $\vec{b}$ is 60°. If $\frac{1}{8}\vec{a}$ is a unit vector, then $\left|\vec{b}\right|$ is equal to :

Open solution

(A) 4 (B) 6 (C) 5 (D) 8

Q25. MCQ (Single Correct Answer)

The function

$f(x) = \left|x^2 - 2x - 3\right| . e^{|9x^2-12x+4|}$ is not differentiable at exactly :

Open solution

(A) four points (B) three points (C) two points (D) one point

Q26. MCQ (Single Correct Answer)

Three numbers are in an increasing geometric progression with common ratio r. If the middle number is doubled, then the new numbers are in an arithmetic progression with common difference d. If the fourth term of GP is 3 r^2, then $r^2 - d$ is equal to :

Open solution

(A) $7 - 7\sqrt{3}$ (B) $7 + \sqrt{3}$ (C) $7 - \sqrt{3}$ (D) $7 + 3\sqrt{3}$

Q27. MCQ (Single Correct Answer)

Which of the following is not correct for relation R on the set of real numbers ?

Open solution

(A) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$ is neither transitive nor symmetric.

(B) $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$ is symmetric and transitive.

(C) $(x, y) \in R \Leftrightarrow |x| - |y| \leq 1$ is reflexive but not symmetric.

(D) $(x, y) \in R \Leftrightarrow |x - y| \leq 1$ is reflexive and symmetric.

Q28. MCQ (Single Correct Answer)

The integral $\int \frac{1}{\sqrt[4]{(x-1)^3(x+2)^5}}\,dx$ is equal to : (where C is a constant of integration)

Open solution

(A) $\frac{3}{4}\left(\frac{x+2}{x-1}\right)^{\frac{1}{4}} + C$ (B) $\frac{3}{4}\left(\frac{x+2}{x-1}\right)^{\frac{5}{4}} + C$

(C) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{\frac{1}{4}} + C$ (D) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{\frac{5}{4}} + C$

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Q29. MCQ (Single Correct Answer)

If p and q are the lengths of the perpendiculars from the origin on the lines,

x cosec α − y sec α = k cot 2α and

x sinα + y cosα = k sin2α

respectively, then k^2 is equal to :

(A) $4p^2 + q^2$ (B) $2p^2 + q^2$ (C) $p^2 + 2q^2$ (D) $p^2 + 4q^2$

Open solution

Q30. MCQ (Single Correct Answer)

cosec18° is a root of the equation :

(A) $x^2 + 2x - 4 = 0$ (B) $4x^2 + 2x - 1 = 0$ (C) $x^2 - 2x + 4 = 0$ (D) $x^2 - 2x - 4 = 0$

Open solution

Q31. MCQ (Single Correct Answer)

If the following system of linear equations

2x + y + z = 5

x − y + z = 3

x + y + az = b

has no solution, then :

(A) $a = -\frac{1}{3}, b \neq \frac{7}{3}$ (B) $a \neq \frac{1}{3}, b = \frac{7}{3}$

(C) $a \neq -\frac{1}{3}, b = \frac{7}{3}$ (D) $a = \frac{1}{3}, b \neq \frac{7}{3}$

Open solution

Q32. MCQ (Single Correct Answer)

The length of the latus rectum of a parabola, whose vertex and focus are on the positive x-axis at a distance R and S (> R) respectively from the origin, is :

(A) 4(S + R) (B) 2(S − R) (C) 4(S − R) (D) 2(S + R)

Open solution

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Q33. MCQ (Single Correct Answer)

If the function

$$f(x)=\begin{cases}\frac{1}{x}\log_e\left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}}\right) & , \quad x<0\\ k & , \quad x=0\\ \frac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+1}-1} & , \quad x>0\end{cases}$$ is continuous

at x = 0, then $\frac{1}{a}+\frac{1}{b}+\frac{4}{k}$ is equal to :

(A) -5 (B) 5 (C) -4 (D) 4

Open solution

Q34. MCQ (Single Correct Answer)

If $\frac{dy}{dx}=\frac{2^{x+y}-2^x}{2^y}$, y(0) = 1, then y(1) is equal to :

(A) $\log_2(2+e)$ (B) $\log_2(1+e)$ (C) $\log_2(2e)$ (D) $\log_2(1+e^2)$

Open solution

Q35. MCQ (Single Correct Answer)

$\lim\limits_{x\to 0}\frac{\sin^2(\pi\cos^4 x)}{x^4}$ is equal to :

(A) π^2 (B) $2\pi^2$ (C) $4\pi^2$ (D) 4π

Open solution

Q36. MCQ (Single Correct Answer)

If $a_r=\cos\frac{2r\pi}{9}+i\sin\frac{2r\pi}{9}$, r = 1, 2, 3, ..., i = $\sqrt{-1}$, then the determinant $\begin{vmatrix}a_1 & a_2 & a_3\\ a_4 & a_5 & a_6\\ a_7 & a_8 & a_9\end{vmatrix}$ is equal to :

(A) $a_2a_6-a_4a_8$ (B) a_9 (C) $a_1a_9-a_3a_7$ (D) a_5

Open solution

Q37. Numerical

Let [t] denote the greatest integer $\le$ t. Then the value of

8. $\int\limits_{-\frac{1}{2}}^{1}([2x]+|x|)dx$ is _____.

Open solution

Q38. Numerical

A point z moves in the complex plane such that $\arg\left(\frac{z-2}{z+2}\right)=\frac{\pi}{4}$, then the minimum value of $\left|z-9\sqrt{2}-2i\right|^2$ is equal to _____.

Open solution

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Q39. Numerical

If 'R' is the least value of 'a' such that the function f(x) = x^2 + ax + 1 is increasing on [1, 2] and 'S' is the greatest value of 'a' such that the function f(x) = x^2 + ax + 1 is decreasing on [1, 2], then the value of |R − S| is _____.

Open solution

Q40. Numerical

The mean of 10 numbers 7 × 8, 10 × 10, 13 × 12, 16 × 14, ... is _____.

Open solution

Q41. Numerical

If the variable line 3x + 4y = α lies between the two
circles $(x-1)^2 + (y-1)^2 = 1$
and $(x-9)^2 + (y-1)^2 = 4$, without intercepting a chord on either circle, then the sum of all the integral values of α is _____.

Open solution

Q42. Numerical

The number of six letter words (with or without meaning), formed using all the letters of the word 'VOWELS', so that all the consonants never come together, is _____.

Open solution

Q43. Numerical

If $x\phi(x) = \int_5^x (3t^2 - 2\phi'(t))dt$, x > −2, and $\phi(0) = 4$, then $\phi(2)$ is _____.

Open solution

Q44. Numerical

If $\left(\frac{3^6}{4^4}\right)k$ is the term, independent of x, in the binomial expansion of $\left(\frac{x}{4} - \frac{12}{x^2}\right)^{12}$, then k is equal to _____.

Open solution

Q45. Numerical

An electric instrument consists of two units. Each unit must function independently for the instrument to operate. The probability that the first unit functions is 0.9 and that of the second unit is 0.8. The instrument is switched on and it fails to operate. If the probability that only the first unit failed and second unit is functioning is p, then 98 p is equal to _____.

Open solution

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Physics 27 questions

Q46. MCQ (Single Correct Answer)

A helicopter is flying horizontally with a speed 'v' at an altitude 'h' has to drop a food packet for a man on the ground. What is the distance of helicopter from the man when the food packet is dropped?

Open solution

(A) $\sqrt{\frac{2ghv^2+1}{h^2}}$

(B) $\sqrt{2ghv^2+h^2}$

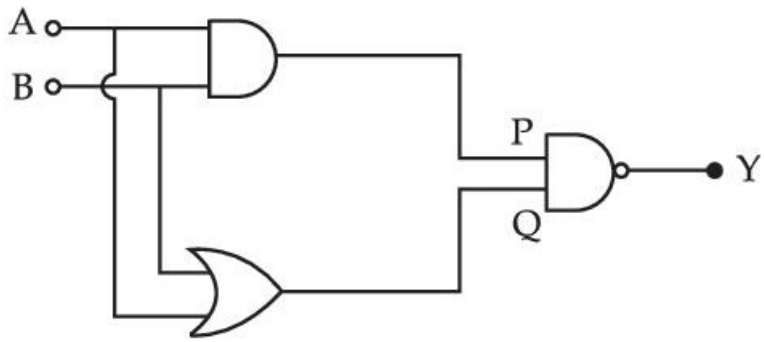
(C) $\sqrt{\frac{2v^2h}{g}+h^2}$

(D) $\sqrt{\frac{2gh}{v^2}+h^2}$

Q47. MCQ (Single Correct Answer)

In the following logic circuit the sequence of the inputs A, B are (0, 0), (0, 1), (1, 0) and (1, 1). The output Y for this sequence will be :

Open solution



(A) 1, 0, 1, 0

(B) 0, 1, 0, 1

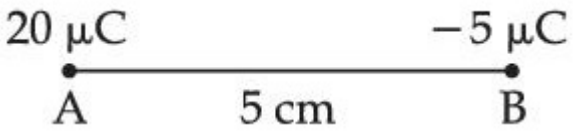
(C) 1, 1, 1, 0

(D) 0, 0, 1, 1

Q48. MCQ (Single Correct Answer)

Two particles A and B having charges $20\mu C$ and $-5\mu C$ respectively are held fixed with a separation of 5 cm. At what position a third charged particle should be placed so that it does not experience a net electric force?

Open solution



(A) At 5 cm from 20 μC on the left side of system

(B) At 5 cm from -5 μC on the right side

(C) At 1.25 cm from -5 μC between two charges

(D) At midpoint between two charges

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Q49. MCQ (Single Correct Answer)

An object is placed at the focus of concave lens having focal length f. What is the magnification and distance of the image from the optical centre of the lens?

(A) 1, ∞ (B) Very high, ∞ (C) $\frac{1}{2}$, $\frac{f}{2}$ (D) $\frac{1}{4}$, $\frac{f}{4}$

Open solution

Q50. MCQ (Single Correct Answer)

A coil having N turns is wound tightly in the form of a spiral with inner and outer radii 'a' and 'b' respectively. Find the magnetic field at centre, when a current I passes through coil:

(A) $\frac{\mu_0 IN}{2(b-a)}\log_e\left(\frac{b}{a}\right)$ (B) $\frac{\mu_0 I}{8}\left[\frac{a+b}{a-b}\right]$

(C) $\frac{\mu_0 I}{4(a-b)}\left[\frac{1}{a}-\frac{1}{b}\right]$ (D) $\frac{\mu_0 I}{8}\left(\frac{a-b}{a+b}\right)$

Open solution

Q51. MCQ (Single Correct Answer)

A body of mass M moving at speed V_0 collides elastically with a mass 'm' at rest. After the collision, the two masses move at angles θ_1 and θ_2 with respect to the initial direction of motion of the body of mass M. The largest possible value of the ratio M/m, for which the angles θ_1 and θ_2 will be equal, is :

(A) 4 (B) 1 (C) 3 (D) 2

Open solution

Q52. MCQ (Single Correct Answer)

The masses and radii of the earth and moon are (M_1, R_1) and (M_2, R_2) respectively. Their centres are at a distance 'r' apart. Find the minimum escape velocity for a particle of mass 'm' to be projected from the middle of these two masses :

(A) $V=\frac{1}{2}\sqrt{\frac{4G(M_1+M_2)}{r}}$ (B) $V=\sqrt{\frac{4G(M_1+M_2)}{r}}$

(C) $V=\frac{1}{2}\sqrt{\frac{2G(M_1+M_2)}{r}}$ (D) $V=\frac{\sqrt{2G}(M_1+M_2)}{r}$

Open solution

Q53. MCQ (Single Correct Answer)

A small square loop of side 'a' and one turn is placed inside a larger square loop of side b and one turn (b >> a). The two loops are coplanar with their centres coinciding. If a current I is passed in the square loop of side 'b', then the coefficient of mutual inductance between the two loops is :

(A) $\frac{\mu_0}{4\pi}8\sqrt{2}\frac{a^2}{b}$ (B) $\frac{\mu_0}{4\pi}\frac{8\sqrt{2}}{a}$

(C) $\frac{\mu_0}{4\pi}8\sqrt{2}\frac{b^2}{a}$ (D) $\frac{\mu_0}{4\pi}\frac{8\sqrt{2}}{b}$

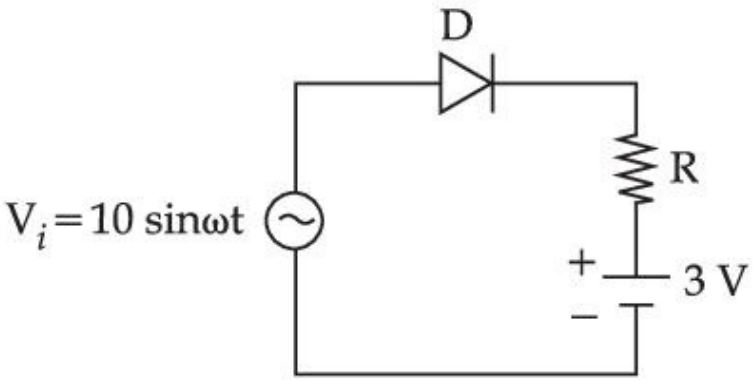
Open solution

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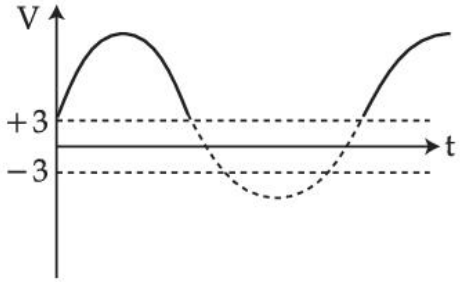
Q54. MCQ (Single Correct Answer)

Open solution

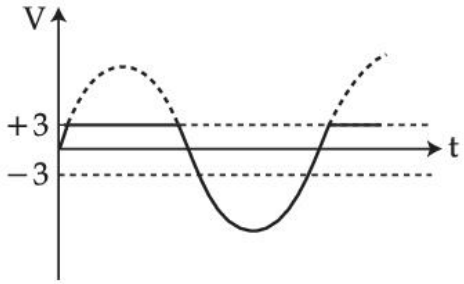
Choose the correct waveform that can represent the voltage across R of the following circuit, assuming the diode is ideal one :



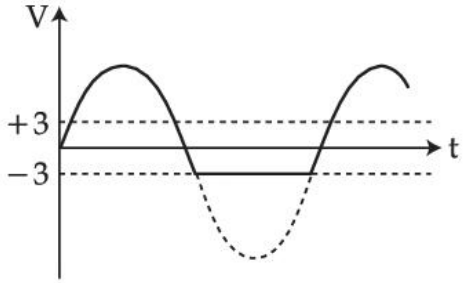
(A)



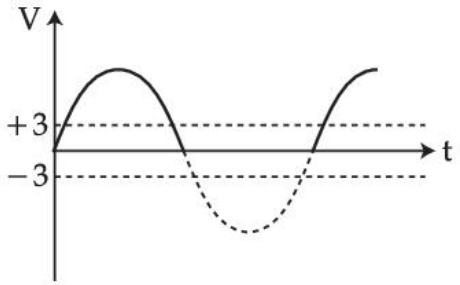
(B)



(C)



(D)



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Q55. MCQ (Single Correct Answer)

Open solution

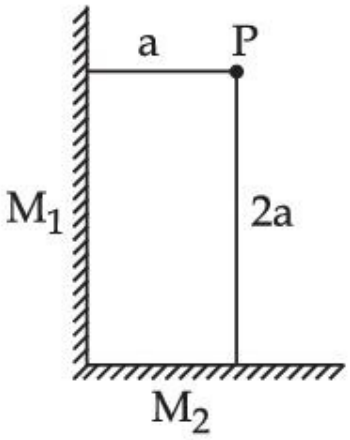
A uniform heavy rod of weight 10 kg ms^{-2}, cross-sectional area 100 cm^2 and length 20 cm is hanging from a fixed support. Young modulus of the material of the rod is 2×10^{11} Nm^{-2}. Neglecting the lateral contraction, find the elongation of rod due to its own weight.

(A) 2×10^{-9} m (B) 5×10^{-8} m (C) 4×10^{-8} m (D) 5×10^{-10} m

Q56. MCQ (Single Correct Answer)

Open solution

Two plane mirrors M_1 and M_2 are at right angle to each other shown. A point source 'P' is placed at 'a' and '2a' meter away from M_1 and M_2 respectively. The shortest distance between the images thus formed is : (Take $\sqrt{5}$ = 2.3)



(A) 3a (B) 4.6a (C) 2.3a (D) $2\sqrt{10}$a

Q57. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

	List - I		List - II
(a)	Torque	(i)	MLT^{-1}
(b)	Impulse	(ii)	MT^{-2}
(c)	Tension	(iii)	ML^2T^{-2}
(d)	Surface Tension	(iv)	MLT^{-2}

Choose the most appropriate answer from the options given below :

(A) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii) (B) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(C) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii) (D) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

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Q58. MCQ (Single Correct Answer)

For an ideal gas the instantaneous change in pressure 'p' with volume 'v' is given by the equation $\frac{dp}{dv} = -ap$. If p = p_0 at v =0 is the given boundary condition, then the maximum temperature one mole of gas can attain is : (Here R is the gas constant)

Open solution

(A) $\frac{p_0}{aeR}$ (B) $\frac{ap_0}{eR}$ (C) infinity (D) 0°C

Q59. MCQ (Single Correct Answer)

Which of the following equations is dimensionally incorrect?

Where t = time, h = height, s = surface tension, θ = angle, ρ = density, a, r = radius, g = acceleration due to gravity, v = volume, p = pressure, W = work done, Γ = torque, $\in$ = permittivity, E = electric field, J = current density, L = length.

Open solution

(A) $v = \frac{\pi p a^4}{8\eta L}$ (B) $h = \frac{2s\cos\theta}{\rho r g}$

(C) $J = \in \frac{\partial E}{\partial t}$ (D) $W = \Gamma\theta$

Q60. MCQ (Single Correct Answer)

Angular momentum of a single particle moving with constant speed along circular path :

Open solution

(A) changes in magnitude but remains same in the direction

(B) remains same in magnitude and direction

(C) remains same in magnitude but changes in the direction

(D) is zero

Q61. MCQ (Single Correct Answer)

In an ac circuit, an inductor, a capacitor and a resistor are connected in series with X_L = R = X_C. Impedance of this circuit is :

Open solution

(A) $2R^2$ (B) Zero (C) R (D) $R\sqrt{2}$

Q62. MCQ (Single Correct Answer)

A moving proton and electron have the same de-Broglie wavelength. If K and P denote the K.E. and momentum respectively. Then choose the correct option :

Open solution

(A) $K_p < K_e$ and $P_p = P_e$ (B) $K_p = K_e$ and $P_p = P_e$

(C) $K_p < K_e$ an $P_p < P_e$ (D) $K_p > K_e$ and $P_p = P_e$

Q63. MCQ (Single Correct Answer)

Consider a galvanometer shunted with 5Ω resistance and 2% of current passes through it. What is the resistance of the given galvanometer ?

Open solution

(A) 300 Ω (B) 344 Ω (C) 245 Ω (D) 226 Ω

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Q64. Numerical

When a rubber ball is taken to a depth of _____m in deep sea, its volume decreases by 0.5%.

(The bulk modulus of rubber = 9.8×10^8 Nm^{-2}, Density of sea water = 10^3 kgm^{-3}, g = 9.8 m/s^2)

Open solution

Q65. Numerical

A particle of mass 1 kg is hanging from a spring of force constant 100 Nm^{-1}. The mass is pulled slightly downward and released so that it executes free simple harmonic motion with time period T. The time when the kinetic energy and potential energy of the system will become equal, is $\frac{T}{x}$. The value of x is _____.

Open solution

Q66. Numerical

A square shaped wire with resistance of each side 3Ω is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be _____.

Open solution

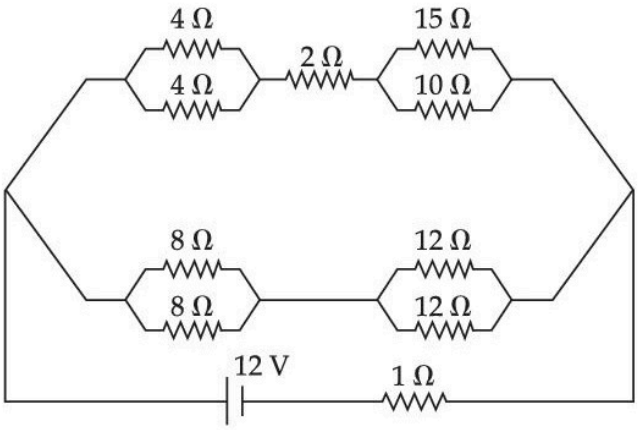
Q67. Numerical

A wire having a linear mass density 9.0×10^{-4} kg/m is stretched between two rigid supports with a tension of 900 N. The wire resonates at a frequency of 500 Hz. The next higher frequency at which the same wire resonates is 550 Hz. The length of the wire is _____ m.

Open solution

Q68. Numerical

The voltage drop across 15Ω resistance in the given figure will be _____ V.



Open solution

Q69. Numerical

A block moving horizontally on a smooth surface with a speed of 40 ms^{-1} splits into two equal parts. If one of the parts moves at 60 ms^{-1} in the same direction, then the fractional change in the kinetic energy will be x : 4 where x = _____.

Open solution

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Q70. Numerical

The electric field in an electromagnetic wave is given by E = (50 NC^{-1}) $\sin\omega$ (t − x/c)

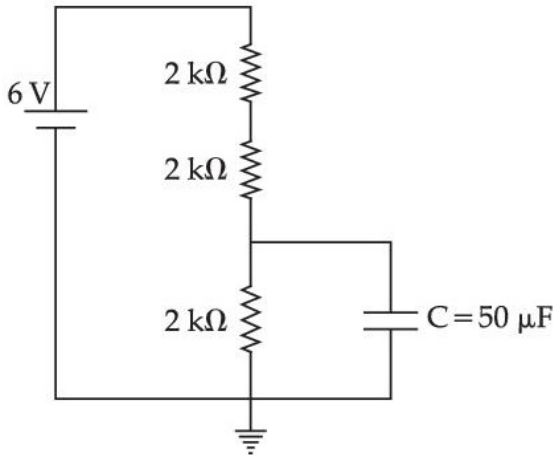
The energy contained in a cylinder of volume V is 5.5 × 10^{-12} J. The value of V is _____ cm^3. (given $\in_0$ = 8.8 × $10^{-12}C^2N^{-1}m^{-2}$)

Open solution

Q71. Numerical

A capacitor of 50 μF is connected in a circuit as shown in figure. The charge on the upper plate of the capacitor is _____μC.

Open solution



Q72. Numerical

A car is moving on a plane inclined at 30° to the horizontal with an acceleration of 10 ms^{-2} parallel to the plane upward. A bob is suspended by a string from the roof of the car. The angle in degrees which the string makes with the vertical is _____. (Take g = 10 ms^{-2})

Open solution

Out of syllabus 18 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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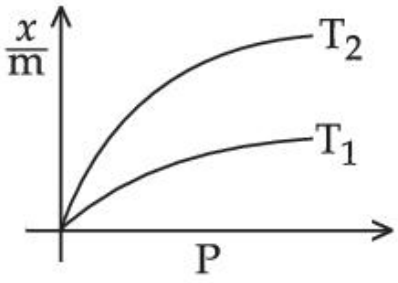
Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

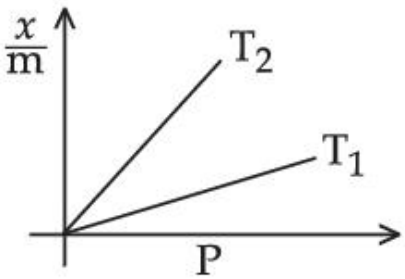
Select the graph that correctly describes the adsorption isotherms at two temperatures T_1, and T_2 ($T_1 > T_2$) for a gas :

(x - mass of the gas absorbed ; m - mass of adsorbent ; P - pressure)

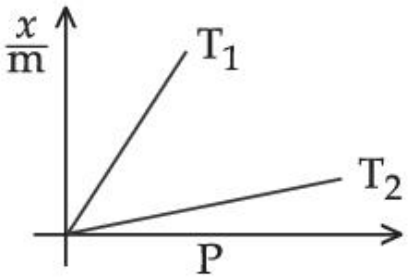
(A)



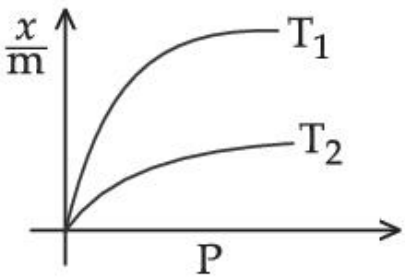
(B)



(C)



(D)



Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The major component/ingredient of Portland Cement is :

(A) tricalcium aluminate

(B) tricalcium silicate

(C) dicalcium aluminate

(D) dicalcium silicate

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Q75. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Aluminium is extracted from bauxite by the electrolysis of molten mixture of Al_2O_3 with cryolite.

Reason (R) : The oxidation state of Al in cryolite is +3.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) (A) is true but (R) is false.

(B) (A) is false but (R) is true.

(C) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(D) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Monomer of Novolac is :

(A) 3-Hydroxybutanoic acid

(B) phenol and melamine

(C) o-Hydroxymethylphenol

(D) 1, 3-Butadiene and styrene

Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements :

Statement I : The process of producing syn-gas is called gasification of coal.

Statement II : The composition of syn-gas is CO + CO_2 + H_2 (1 : 1 : 1)

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

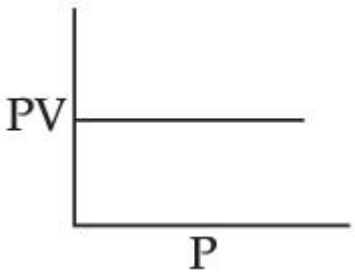
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Q78. MCQ (Single Correct Answer) · **Chemistry**

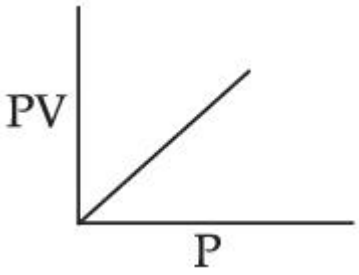
Which one of the following is the correct PV vs P plot at constant temperature for an ideal gas? (P and V stand for pressure and volume of the gas respectively)

Open solution

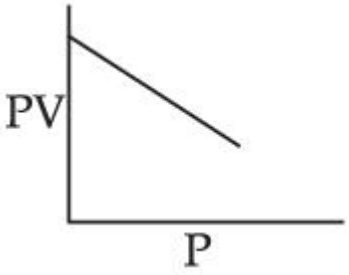
(A)



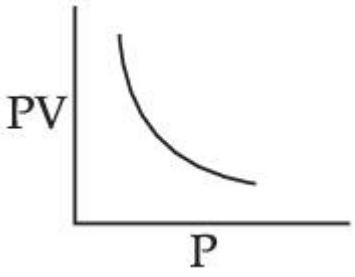
(B)



(C)



(D)



Q79. MCQ (Single Correct Answer) · **Chemistry**

BOD values (in ppm) for clean water (A) and polluted water (B) are expected respectively :

(A) A > 50, B < 27 (B) A > 25, B < 17 (C) A < 5, B > 17 (D) A > 15, B > 47

Open solution

Q80. Numerical · **Chemistry**

Consider the sulphides HgS, PbS, CuS, Sb_2S_3, As_2S_3 and CdS. Number of these sulphides soluble in 50% HNO_3 is _____.

Open solution

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Q81. Numerical · **Chemistry**

The number of halogen/(s) forming halic (V) acid is _____.

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let *, $\square \in \{\wedge, \vee\}$ be such that the Boolean expression $(p * \sim q) \Rightarrow (p \square q)$ is a tautology. Then :

(A) * = $\vee$, $\square$ = $\vee$

(B) * = $\wedge$, $\square$ = $\wedge$

(C) * = $\wedge$, $\square$ = $\vee$

(D) * = $\vee$, $\square$ = $\wedge$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

The sum of 10 terms of the series

$\frac{3}{1^2\times2^2} + \frac{5}{2^2\times3^2} + \frac{7}{3^2\times4^2} + \ldots$. is :

(A) 1 (B) $\frac{120}{121}$ (C) $\frac{99}{100}$ (D) $\frac{143}{144}$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Let the equation of the plane, that passes through the point (1, 4, -3) and contains the line of intersection of the
planes $3x - 2y + 4z - 7 = 0$
and $x + 5y - 2z + 9 = 0$, be
$\alpha x + \beta y + \gamma z + 3 = 0$, then $\alpha + \beta + \gamma$ is equal to :

(A) -23 (B) -15 (C) 23 (D) 15

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

A vertical pole fixed to the horizontal ground is divided in the ratio 3 : 7 by a mark on it with lower part shorter than the upper part. If the two parts subtend equal angles at a point on the ground 18 m away from the base of the pole, then the height of the pole (in meters) is :

(A) $12\sqrt{15}$ (B) $12\sqrt{10}$ (C) $8\sqrt{10}$ (D) $6\sqrt{10}$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

The line $12x\cos\theta + 5y\sin\theta = 60$ is tangent to which of the following curves?

(A) $x^2 + y^2 = 169$ (B) $144x^2 + 25y^2 = 3600$

(C) $25x^2 + 12y^2 = 3600$ (D) $x^2 + y^2 = 60$

Open solution

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Q87. Numerical · **Mathematics**

The square of the distance of the point of intersection

of the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$ and the plane $2x - y + z = 6$ from the point $(-1, -1, 2)$ is _____.

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

A reversible engine has an efficiency of $\frac{1}{4}$. If the temperature of the sink is reduced by 58°C, its efficiency becomes double. Calculate the temperature of the sink :

(A) 174°C (B) 280°C (C) 180.4°C (D) 382°C

Open solution

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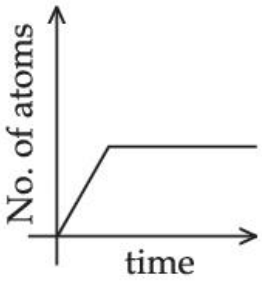
Q89. MCQ (Single Correct Answer) · **Physics**

Open solution

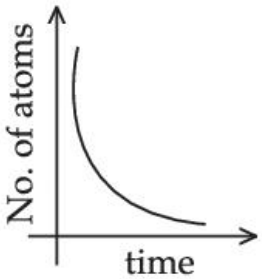
A sample of a radioactive nucleus A disintegrates to another radioactive nucleus B, which in turn disintegrates to some other stable nucleus C. Plot of a graph showing the variation of number of atoms of nucleus B versus time is :

(Assume that at t = 0, there are no B atoms in the sample)

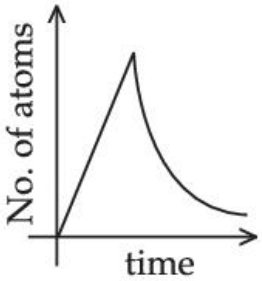
(A)



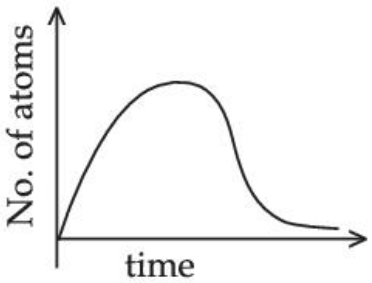
(B)



(C)



(D)



Q90. Numerical · **Physics**

Open solution

If the sum of the heights of transmitting and receiving antennas in the line of sight of communication is fixed at 160 m, then the maximum range of LOS communication is _____ km. (Take radius of Earth = 6400 km)

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Answer Key

Chemistry

1. **D**	2. **B**	3. **A**	4. **B**	5. **B**
6. **A**	7. **C**	8. **A**	9. **B**	10. **C**
11. **D**	12. **D**	13. **A**	14. **25**	15. **20**
16. **5**	17. **2**	18. **1**	19. **45**	20. **7**
21. **108**				

Mathematics

22. **C**	23. **D**	24. **C**	25. **C**	26. **B**
27. **B**	28. **C**	29. **A**	30. **D**	31. **D**
32. **C**	33. **A**	34. **B**	35. **C**	36. **C**
37. **5**	38. **98**	39. **2**	40. **398**	41. **165**
42. **576**	43. **4**	44. **55**	45. **28**	

Physics

46. **C**	47. **C**	48. **B**	49. **C**	50. **A**
51. **C**	52. **B**	53. **A**	54. **D**	55. **D**
56. **B**	57. **A**	58. **A**	59. **A**	60. **B**
61. **C**	62. **A**	63. **C**	64. **500**	65. **8**
66. **3**	67. **10**	68. **6**	69. **1**	70. **500**
71. **100**	72. **30**			

Out of syllabus

73. **A**	74. **B**	75. **D**	76. **C**	77. **B**
78. **A**	79. **C**	80. **4**	81. **3**	82. **C**
83. **B**	84. **A**	85. **B**	86. **B**	87. **61**
88. **A**	89. **D**	90. **64**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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